6 Discussion

6.1 Bridging Online and Offline Merging via Step-K Online Merging Optimizers

Offline Model Merging occurs after RLHF training concludes, combining the trained model with the SFT model. In contrast, Online Model Merging merges models at each training step. To bridge the gap, we propose step-K online merging optimizers, as detailed in Alg. 2. In step-K online merging, the optimizer caches accumulated update Δ_c, which is zero at the beginning of the training. Then, it applies the standard Adam update to the parameters for K optimization steps. At the K-th step, it first reverts the parameters to their cached state from the previous K-th step and calculates the delta parameters as the difference between current and cached parameters. These delta parameters are then merged with those of the reference model. The optimizer then updates the current parameters with the merged ones and refreshes the cached parameters. The online gap step K controls the **onlineness** for the step-K online merging optimizers. The pseudo-codes are detailed in Alg. 2. When K equals 1, the step-K online merging optimizer functions as the online merging optimizer introduced in §4; When K equals the total number of training steps, implying a single merge at the end of the training, the step-K online merging optimizer operates as the offline merging optimizer described in §3.

We then examine how the onlineness may affect the overall performance of step-K online merging optimizers. We run step-K optimizers on UltraFeedback trained from Qwen1.5-1.8B-Chat with online gap steps in $\{5, 10, 50, 100, 200\}$, whose results are shown in Fig. 3. As the online merging gap step increases, we witness a significant drop in MT-Bench scores for both OnDARE and OnTIES, with scores gradually approaching the performance of AdamW. The decline in average benchmark scores is less pronounced in MT-Bench, indicating that the "onlineness" of model merging is more critical for achieving higher rewards than for mitigating forgetting.

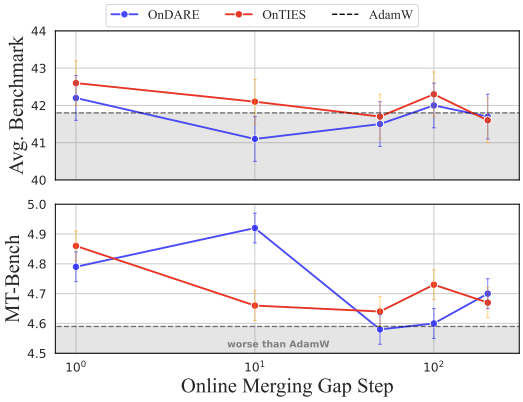

Figure 3: Analysis of Online Merging Gap Step

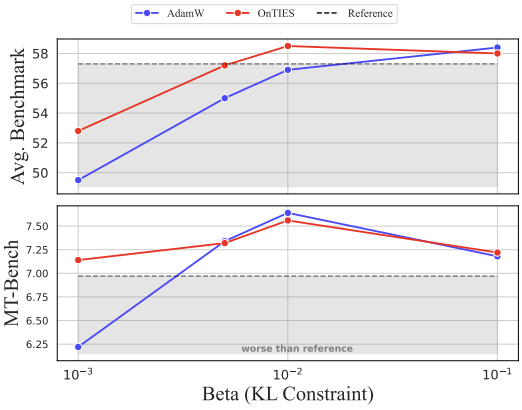

Figure 4: Analysis of KL constraints

6.2 Complementary Effect of KL Constraints and Online Merging

We further analyze the complementary effect of widely used KL constraint and online merging. We conduct experiments on Qwen1.5-7B-Chat with AdamW and OnTIES under extremely small β of the DPO algorithm, indicating very few KL constraints on the training. The results are shown in Fig. 4. At $\beta = 1e^{-1}$, AdamW and OnTIES show similar performance on both the MT-Bench and the average benchmark scores. However, as the β decreases under $1e^{-1}$, OnTIES is significantly better than AdamW on the average benchmark score, showing regularization from the online merging optimizer mitigates forgetting under lower KL constraints. Furthermore, OnTIES still achieves MT-Bench over 6.9 (MT-Bench of the reference SFT model) under an extremely small β ($1e^{-3}$), while AdamW falls to 6.25, demonstrating the complementary effect of KL constraints and online merging.

7 Conclusion

In this work, we investigate the issue of alignment tax in the RLHF training of LLMs. Drawing inspiration from offline merging methods, we develop online merging optimizers that integrate general task arithmetic merging into each RLHF optimization step. Extensive experiments conducted on

various backbone LLMs demonstrate that online merging optimizers better mitigate alignment tax and achieve superior alignment performance compared to regularization and offline merging baselines. Additionally, we propose step-K online merging optimizers to bridge the gap between online and offline merging, providing in-depth analyses on the effects of hyper-parameters and ablations. While our primary focus is on using online merging optimizers for LLM alignment, their applications can naturally extend to other domains facing catastrophic forgetting, such as the continual learning of LLMs. We hope that online merging optimizers will become a representative optimizer in future RLHF training, enabling researchers to more boldly pursue reward maximization without concerns about model degradation, ultimately producing more helpful, honest, and harmless LLMs.

Limitations

Major limitations of online merging optimizers are related to the parameter efficiency. Online merging optimizers enhance the memory requirements as they require a cache of additional delta parameters of the reference model as the counterpart of the delta updated weights in training. At the same time, they can not be applied to the LoRA training, except for the cases where the reference models are also trained with LoRA adapters. However, this limitation can be further eliminated by applying GaLore (Zhao et al., 2024) with the online merging optimizers.

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